CGOcc: Learning Cross-Gaussian Dependencies for 3D Semantic Occupancy Prediction

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Abstract

*In Gaussian-based semantic occupancy prediction, overlapping primitives can provide ambiguous class predictions, making their joint interpretation important for voxel classification. However, current methods using spatially weighted mixtures neglect explicit modeling of cross-primitive dependencies, limiting the readout's ability to utilize category-specific relational evidence. Hence, we introduce **CGOcc**, a supervised multi-camera framework that explicitly models **C**ross-**G**aussian dependencies for 3D semantic **Occ**upancy prediction. CGOcc integrates appearance and geometric observations to refine Gaussian states and encodes cross-Gaussian dependencies in a diagonal-plus-low-rank representation. To incorporate these dependencies into voxel classification, CGOcc uses a class-conditional projection to extract diagonal and signed cross-Gaussian statistics without constructing a dense dependency matrix. The readout then refines occupied-class probabilities using these statistics while preserving a learned reference's occupied mass and free probability. Extensive experiments on nuScenes demonstrate CGOcc achieves state-of-the-art performance, with larger gains at high Gaussian overlap and improved predictions near boundaries. Ablation studies validate the effectiveness of covariance-aware prediction and support its synergy with geometric interaction.*

1. Introduction

Camera-based 3D semantic occupancy prediction describes scene geometry and semantics beyond bounding boxes for autonomous perception [40, 44]. Alongside dense volumetric and tri-perspective representations [14, 44], semantic Gaussians offer adaptive, continuous scene representations [15, 16]. Their overlapping supports make voxel semantics depend on how multiple primitive predictions are combined.

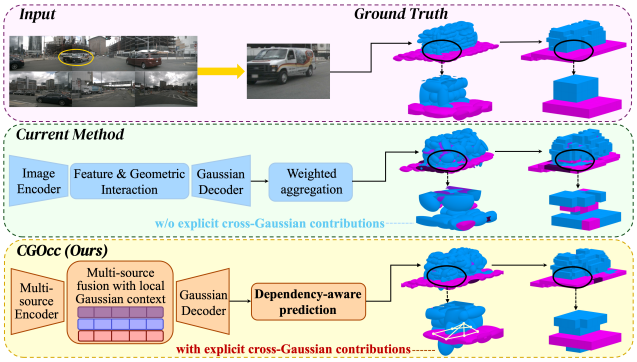


Figure 1. **Cross-Gaussian dependencies for semantic occupancy.** A schematic vehicle–road example compares weighted aggregation (middle) with CGOcc (bottom), with input and ground truth above. Blue and magenta denote vehicle and road; yellow and black circles mark the input region and enlarged regions, respectively. Pale yellow edges indicate selected cross-Gaussian dependencies.

Foundation models provide complementary priors: DINOv2 supplies visual features [33], Metric3Dv2 metric depth [12], and VGGT multi-view geometry [42]. GaussTR aligns rendered features and depth with foundation-model outputs [17]. These observations may vary in usefulness across views and locations. Refinement therefore calls for integrating heterogeneous observations according to their geometric relationships to each Gaussian and incorporating local context from neighboring primitives.

At semantic readout, GaussianFormer-2 computes spatially weighted mixtures of individual class probabilities [16]. Although decoder interactions can influence in-

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dividual Gaussian predictions, the mixture lacks an explicit class-conditioned cross-Gaussian term. At fixed spatial weights and primitive class probabilities, different cross-Gaussian relationships yield the same semantic mixture. This limits the readout's ability to utilize class-specific relational evidence to resolve ambiguity among overlapping primitives, as illustrated in Fig. 1.

We introduce **CGOcc**, a supervised multi-camera framework that explicitly models **C**ross-**G**aussian dependencies for 3D semantic **Occ**upancy prediction. Learned dependencies provide class-conditional readout corrections that can complement ambiguous individual predictions. Diagonal statistics summarize individual contributions, while cross statistics retain signed interactions between primitives contributing to the same class and voxel. Structured segmentation and factorization models offer principles for joint variation and predictive interactions [11, 30, 38]. Here, covariance parameterizes task-learned Gaussian dependencies, distinct from each primitive's spatial covariance.

To connect observation integration with dependency modeling, the interaction module integrates observations sampled around evolving Gaussians through cross-source exchange, geometry-biased query attention, and local Gaussian-neighbor messages. The prediction module learns diagonal-plus-low-rank dependencies from the resulting states and observations. Spatial weights and class probabilities project this representation into diagonal and signed cross-Gaussian statistics, restricting cross contributions to shared support without enumerating primitive pairs or constructing a dense dependency matrix. A learned reference uses context and diagonal statistics; bounded corrections driven by the explicit cross input reweight its occupied-class probabilities. This refinement preserves the reference's occupied mass and free probability, and exactly recovers the reference when the cross input vanishes.

Extensive experiments on nuScenes show that CGOcc achieves the best performance among the compared methods. In spatial analyses, mIoU gains increase across high-overlap bins and peak in the highest-overlap bin; near boundaries, the largest margin occurs at 0.4 m from ground-truth interfaces. The full class-conditional readout outperforms its ablated variants, and paired ablations support conditional synergy between decoder mechanisms and the covariance-aware readout.

In summary, Our main contributions are as follows:

- A geometry-conditioned multi-source interaction module that integrates heterogeneous observations around evolving Gaussian states and propagates local Gaussian context.
- A class-conditional dependency projection and mass-preserving semantic readout that explicitly incorporate cross-Gaussian statistics into voxel classification without constructing a dense dependency matrix.
- Empirical evidence that the proposed readout improves over ablated variants, together with synergy between decoder mechanisms and covariance-aware prediction, larger gains at high Gaussian overlap, and improved semantic predictions near boundaries.

2. Related Work

Gaussian Representations for 3D Semantic Occupancy. The transition from scene-specific Gaussian rendering [18] to semantic voxel occupancy [15] connects representation design with rendering-based supervision. On the representation side, methods separate occupancy from semantic mixtures [16], combine depth-guided initialization with decoupled geometric and semantic aggregation [35], and incorporate scene evolution, spatial-temporal fusion, and incremental memory [47, 49, 60]. Rendering provides self- or weak supervision [2, 9, 39], including GaussTR's foundation-model alignment of rendered features and depth for open-vocabulary inference [17]. GaussRender extends this supervisory role to architectures without Gaussian internal representations through projective supervision [5]. Here, spatial covariance describes individual Gaussian shape, leaving cross-primitive semantic dependencies as a separate modeling question.

Foundation Model Priors and Gaussian Feature Interaction. DINOv2, Metric3Dv2, and VGGT supply complementary visual features, metric depth and normals, and multi-view geometry [12, 33, 42]. While Feature 3DGS and LangSplat transfer visual and language features into scene-level representations [36, 58], feed-forward occupancy methods adapt frozen geometric features and extend surface priors into volumetric support [50, 56], with further integration of ray geometry and image context [28, 57]. Their integration depends on how observations interact and contribute: Set Transformer models observation interactions [24], deformable attention localizes sampling [59], and VALA uses rendering visibility for language aggregation [43]. Our decoder combines appearance and geometric priors with multi-view evidence and local Gaussian context.

Structured Dependencies and Covariance-Aware Semantic Aggregation. Structured prediction captures pairwise label compatibility [22, 55], spatially correlated aleatoric uncertainty through Gaussian logits with diagonal-plus-low-rank covariance [30], and segmentation ambiguity or correlated annotation errors [7, 21, 32], whereas factorization and bilinear models learn predictive feature interactions [10, 11, 27, 38, 48]. For semantic mapping, Bayesian kernel inference and ConvBKI fuse spatial evidence [8, 45], with E2-BKI adding evidential uncertainty [19]. Continuous Semantic Splatting maintains per-ellipsoid Dirichlet semantic beliefs, but computes rendered variance using squared weights under independence, excluding cross-ellipsoid covariance [46]. Motivated by this distinction,

our method projects learned cross-Gaussian dependencies through spatial support and class probabilities, using the resulting class-conditional statistics as discriminative signals to refine voxel-level semantic predictions.

3. Methodology

This section presents a comprehensive elaboration of the CGOcc framework. We initialize Gaussian centers from depth (Sec. 3.1) and refine their states using multi-view appearance and geometric features (Sec. 3.2). We then model dependencies between Gaussians and project them into class-specific statistics for voxel semantic prediction (Sec. 3.3). Voxel and auxiliary Gaussian supervision jointly train the task-specific modules (Sec. 3.4). An overview of the CGOcc framework is depicted in Fig. 2.

3.1. Problem Formulation and Gaussian Representation

Given surround-view RGB images $\{I_m, \mathbf{K}_m, \mathbf{T}_m\}_{m=1}^{M}$ with intrinsics $\mathbf{K}_m$ and camera-to-ego extrinsics $\mathbf{T}_m$, we predict voxel distributions $\mathbf{P} \in [0,1]^{X\times Y\times Z\times(C+1)}$ over $\mathcal{C} = \mathcal{C}_{\text{occ}} \cup \{\text{free}\}$. Here $\mathcal{C}_{\text{occ}}$ contains C occupied classes. Voxel labels $\mathbf{Y} \in \mathcal{C}^{X\times Y\times Z}$ supervise predictions on an ego-frame grid with centers $\mathbf{x}_v$, $v \in \mathcal{V}$, and predictions $\hat{Y}_v = \arg\max_{c\in\mathcal{C}} P_{v,c}$.

We represent the scene with N anisotropic Gaussians [15], each with center $\boldsymbol{\mu}_g \in \mathbb{R}^3$, opacity $o_g \in [0,1]$, query $\mathbf{q}_g \in \mathbb{R}^d$, and spatial covariance $\boldsymbol{\Sigma}_g = \mathbf{R}_g \operatorname{diag}(\mathbf{s}_g^2)\mathbf{R}_g^\top$, where $\mathbf{s}_g > 0$ contains the scales and $\mathbf{R}_g$ is the rotation of a unit quaternion. A shared head predicts occupied-class probabilities $\mathbf{p}_g = \operatorname{softmax}(f_{\text{sem}}(\mathbf{q}_g))$, while free space is inferred at the voxel level. For $\mathbf{d}_{vg} = \mathbf{x}_v - \boldsymbol{\mu}_g$, each Gaussian contributes

$$\rho_{vg} = o_g \exp\!\left(-\tfrac{1}{2}\mathbf{d}_{vg}^\top \boldsymbol{\Sigma}_g^{-1}\mathbf{d}_{vg}\right) \tag{1}$$

within its splatting neighborhood.

Frozen DINOv2 [33] with GaussTR's frozen pyramid neck [17] provides appearance features, while frozen VGGT [42] jointly extracts geometric features from the views. Following GaussTR, we initialize Gaussian centers by calibrated back-projection of Metric3D v2 depth [12] at fixed image references. The resulting states are refined through the interaction blocks below.

3.2. Geometry-Conditioned Multi-Source Gaussian Interaction

Appearance and geometric observations provide complementary cues for Gaussian refinement, but their usefulness can vary across views and locations. CGOcc therefore allows these observations to exchange information and exploits camera-relative geometry to guide their aggregation into each Gaussian query. Refinement proceeds through two successive blocks, with neighbor messages added in the second to complement each Gaussian's sampled observations with local spatial context.

Local observation construction. In block $\ell \in \{0,1\}$, we sample features at $\mathbf{c}_g^{(\ell)} = \boldsymbol{\mu}_g^{(\ell)} + \boldsymbol{\delta}_{g,\text{read}}^{(\ell)}$ and $\mathbf{c}_g^{(\ell)} + \Delta\mathbf{x}_g^{(\ell)}$, where the additional offset is predicted from the query and scaled by the current Gaussian size [59]. Calibrated bilinear sampling and source-specific residual adapters [6] map valid image features into a shared d-dimensional space. Appearance levels are fused with learned masked-softmax weights, while three VGGT levels remain separate. Together with camera, source, and sampling embeddings, these features form the observation set $\mathbf{O}_g^{(\ell)} \in \mathbb{R}^{S\times d}$ with valid indices $\mathcal{J}_g^{(\ell)}$ and $S = 2M(1+3)$.

Geometry-conditioned observation interaction. We apply masked self-attention across observations [24] and, inspired by attention-output gating [37], use observation-conditioned residual gating to modulate the resulting updates before query aggregation. Omitting block indices, we denote the projected self-attention output from $\text{LN}(\mathbf{O}_g)$ by $\mathbf{A}_g$ and write the gated residual update as

$$\mathbf{H}_g = \mathbf{O}_g + \left[\mathbf{1} + \gamma \tanh(f_{\text{gate}}(\text{LN}(\mathbf{O}_g)))\right] \odot \mathbf{A}_g, \tag{2}$$

where γ is learned. The query then attends to $\mathbf{H}_g$ using geometry-biased cross-attention [51]. A learned bounded mapping of camera-relative geometry, source membership, and sampling metadata supplies the head-specific bias b_{gj}^h. With head-specific projections of normalized $\mathbf{q}_g$ and $\mathbf{H}_g$, the update is

$$\begin{aligned} a_{gj}^h &= \operatorname{softmax}_{j\in\mathcal{J}_g}\left(\frac{\langle \mathbf{q}_g^h, \mathbf{H}_{gj}^{h,\text{k}}\rangle}{\sqrt{d_h}} + b_{gj}^h\right), \\ \mathbf{u}_g &= \mathbf{q}_g + W_{\text{out}} \operatorname{Concat}_h\left(\sum_{j\in\mathcal{J}_g} a_{gj}^h \mathbf{H}_{gj}^{h,\text{v}}\right). \end{aligned} \tag{3}$$

where d_h is the head dimension. A feed-forward residual produces $\tilde{\mathbf{q}}_g^{(\ell)}$; all attention aggregates valid observations only.

Gaussian interaction and refinement. We defer graph attention [41, 54] to the second block so that local message passing builds on an initial round of Gaussian refinement from multi-view observations. The first block sets $\mathbf{q}_g^{(1)} = \tilde{\mathbf{q}}_g^{(0)}$, whereas the second applies local graph attention to obtain $\mathbf{q}_g^{(2)}$. Each Gaussian receives messages from up to K nearest initially valid non-self neighbors within radius R, with relative positions conditioning attention and positional messages. After each block, shared heads decode class probabilities and geometric updates from the resulting queries. Cumulative center displacement is bounded relative to initialization, and the updated geometry guides feature sampling in the next block. The final states are used for semantic readout (Sec. 3.3), while intermediate predic-

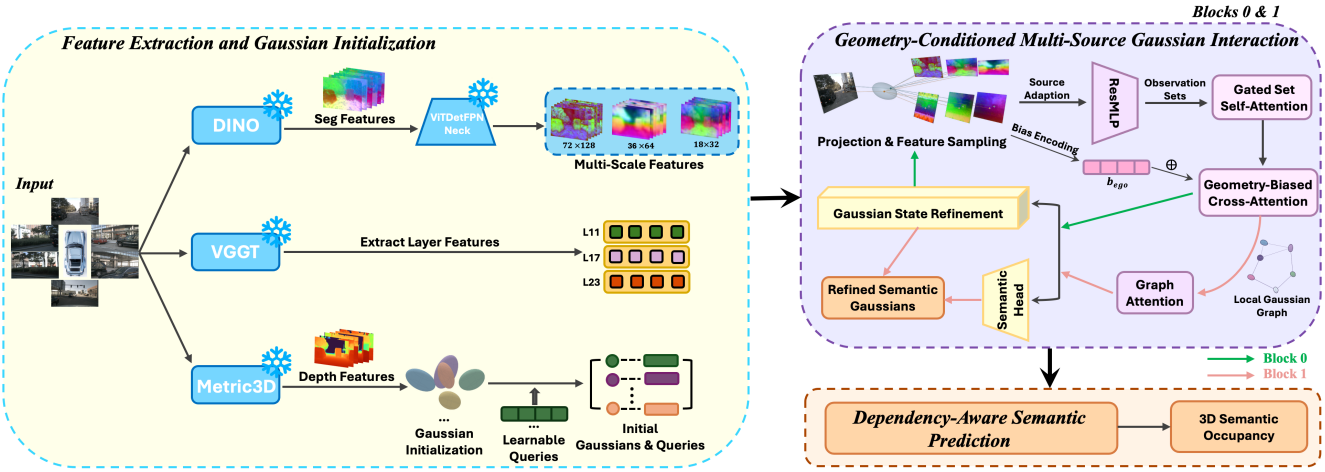


Figure 2. **Overview of CGOcc.** Depth-initialized Gaussians undergo two geometry-conditioned multi-source interaction blocks before dependency-aware semantic readout (Fig. 3). Only Block 1 uses local graph attention; snowflakes denote frozen modules.

tions provide auxiliary supervision without feeding back into subsequent queries.

3.3. Dependency-Aware Semantic Prediction

At voxel readout, however, a spatially weighted mixture of these predictions contains no explicit cross-Gaussian dependency term. CGOcc therefore projects learned dependencies through spatial weights and class probabilities to obtain class-conditional voxel statistics. These statistics guide bounded semantic corrections to a learned reference while preserving its occupied mass and free probability (Fig. 3).

Initial Gaussian-to-voxel aggregation. Following Gaussian-based readout [17, 18, 50], we truncate Eq. (1) to finite splatting neighborhoods, obtaining contributions ω_{vg}. Their sum gives density S_v; normalization by S_v gives weights w_{vg}, set to zero at unsupported voxels. Utilizing the occupancy–semantics decomposition in GaussianFormer-2 [16], we compute

$$\bar{p}_{vc}^{(0)} = \frac{\sum_g \omega_{vg} p_{gc} + \alpha \pi_c}{S_v + \alpha},$$
$$\mathbf{P}_v^{(0)} = [\rho_v \bar{\mathbf{p}}_v^{(0)},\ 1 - \rho_v].$$

The learned prior $\boldsymbol{\pi}$ smooths semantics with strength $\alpha > 0$, including unsupported voxels. A sigmoid of learned log-density calibration plus a depth-derived bias gives occupied probability ρ_v.

Cross-Gaussian dependency modeling. A shared encoder maps Gaussian geometry, opacity, semantics, and a terminal observation summary to $\mathbf{h}_g$. Heads parameterize a

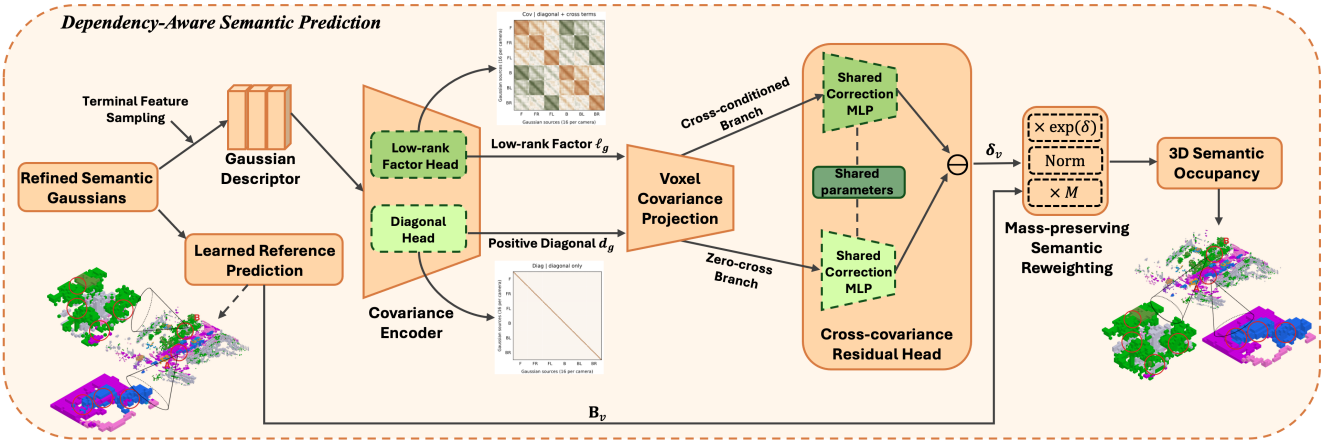


Figure 3. **Overview of Dependency-Aware Semantic Prediction.** Class-conditional projection extracts diagonal and cross-Gaussian statistics from a diagonal-plus-low-rank representation. Shared-head differencing guides semantic reweighting of the learned reference, preserving its occupied mass and free probability.

diagonal-plus-low-rank kernel [30]:

$$\ell_g = \tanh(f_\ell(\mathbf{h}_g)), \qquad d_g = \mathrm{softplus}(f_d(\mathbf{h}_g)) + d_{\min}, \\ K = \mathrm{diag}(\mathbf{d}) + LL^\top. \tag{4}$$

Here $L_{g,:} = \ell_g^\top$, $\ell_g \in \mathbb{R}^r$, and $d_{\min} > 0$. The positive-semidefinite $K \in \mathbb{R}^{N\times N}$ models task-learned dependencies, not spatial shape or calibrated error covariance.

Class-conditional statistical projection. With $a_{vgc} = w_{vg} p_{gc}$ and $\mathbf{a}_{vc} = (a_{v1c}, \dots, a_{vNc})^\top$, projection separates individual and signed cross-Gaussian contributions as $\mathbf{a}_{vc}^\top K \mathbf{a}_{vc} = D_{vc} + X_{vc}$, where:

$$D_{vc} = \sum_g a_{vgc}^2 (d_g + \|\ell_g\|_2^2), \\ X_{vc} = \Big\| \sum_g a_{vgc} \ell_g \Big\|_2^2 - \sum_g a_{vgc}^2 \|\ell_g\|_2^2. \tag{5}$$

Factorization [11, 38] avoids dense K and pair enumeration. For $\epsilon_X > 0$, readout uses normalized descriptors:

$$m_{vc} = \frac{D_{vc}}{1 + D_{vc}}, \qquad \xi_{vc} = \frac{X_{vc}}{D_{vc} + \epsilon_X + |X_{vc}|}.$$

Context $\mathbf{z}_v$ uses spatially weighted pooling [52]. We also define the effective support as $n_v^{\text{eff}} = (\sum_g w_{vg}^2)^{-1}$. For unsupported voxels, pooled statistics and effective support are set to zero.

Mass-preserving semantic correction. The reference $\mathbf{B}_v = F_B(\mathbf{P}_v^{(0)}, \mathbf{z}_v^B, \mathbf{m}_v^B)$ uses separate encoder and factor heads (superscript B). Its context $\mathbf{z}_v^B$ and diagonal descriptor $\mathbf{m}_v^B$ produce bounded semantic and occupancy adjustments $\mathbf{s}_v^B$ and b_v^B. Without explicit cross input, it produces

$$\widehat{\rho}_v = \frac{\rho_v e^{b_v^B}}{\rho_v e^{b_v^B} + 1 - \rho_v}, \\ \mathbf{B}_v = \big[\widehat{\rho}_v \, \mathrm{softmax}(\log \bar{\mathbf{p}}_v^{(0)} + \mathbf{s}_v^B),\ 1 - \widehat{\rho}_v\big]. \tag{6}$$

Writing $H_v(\boldsymbol{\xi}) = H_\psi(\mathbf{z}_v, \mathbf{m}_v, \boldsymbol{\xi})$, we compute:

$$\mathbf{r}_v = H_v(\boldsymbol{\xi}_v) - H_v(\mathbf{0}), \qquad \boldsymbol{\delta}_v = \frac{\tau}{2} \, \mathrm{ctr}\left[\tanh\left(\frac{2\mathbf{r}_v}{\tau}\right)\right]. \tag{7}$$

Both evaluations share parameters, context, and diagonal statistics; $H_v(\mathbf{0})$ centers the correction and is distinct from $\mathbf{B}_v$. Here ctr subtracts the occupied-class mean, and $\tau > 0$ bounds correction magnitudes componentwise. With reference occupied mass $M_v = \sum_{k\in\mathcal{C}_{\text{occ}}} B_{vk} > 0$, normalized exponential reweighting [20] applies to $c \in \mathcal{C}_{\text{occ}}$ when $S_v > 0$ and $n_v^{\text{eff}} > 1$ (up to numerical tolerances):

$$P_{vc} = M_v \frac{B_{vc} e^{\delta_{vc}}}{\sum_{k\in\mathcal{C}_{\text{occ}}} B_{vk} e^{\delta_{vk}}}, \qquad P_{v,\text{free}} = B_{v,\text{free}}. \tag{8}$$

Otherwise, we retain $\mathbf{B}_v$. The update preserves occupied mass and free probability; zero cross input gives zero correction and exactly recovers $\mathbf{B}_v$.

3.4. Training Objective

With feature extractors frozen, voxel and auxiliary Gaussian losses jointly supervise final predictions and primitive states before aggregation. Final voxel predictions receive class-weighted negative log-likelihood $\mathcal{L}_{\text{NLL}}$ and Lovász–Softmax loss $\mathcal{L}_{\text{Lov}}$ [1] on camera-visible voxels, with the latter averaged over all occupied classes. At terminal Gaussian centers, auxiliary supervision combines class-weighted semantic cross-entropy and occupancy binary cross-entropy as

$$\mathcal{L}_{\text{G-aux}} = \mathcal{L}_{\text{G-sem}}^{\text{CE}} + \lambda_{\text{occ}} \mathcal{L}_{\text{G-occ}}^{\text{BCE}}. \tag{9}$$

It uses initially valid, in-volume Gaussians without visibility filtering; its semantic term excludes free-space targets.

We also apply class-weighted semantic cross-entropy $\mathcal{L}_{\text{mid}}$ to the first block for deep supervision [23]. Targets are sampled at detached updated centers and restricted to initially valid, in-volume Gaussians with camera-visible occupied labels, so target assignment provides no gradient to the centers. The overall objective is

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \lambda_{\text{Lov}} \mathcal{L}_{\text{Lov}} + \lambda_{\text{aux}} \mathcal{L}_{\text{G-aux}} + \lambda_{\text{mid}} \mathcal{L}_{\text{mid}}, \tag{10}$$

where the λ coefficients balance the loss terms.

4. Experiment

4.1. Dataset and Metrics

We evaluate our method on Occ3D-nuScenes [40], which provides dense semantic occupancy annotations for nuScenes [3]. Each input comprises RGB images from six surround-view cameras. The prediction volume spans $[-40, 40]$ m along both horizontal axes and $[-1, 5.4]$ m vertically, discretized into a $200 \times 200 \times 16$ grid with a voxel size of 0.4 m. The label space contains 17 occupied semantic classes and one free-space class. Evaluation is restricted to voxels marked as camera-visible by the provided visibility mask [40]. We report mean intersection over union (mIoU) for semantic occupancy and class-agnostic IoU for geometric occupancy, as commonly used in semantic scene completion [4]. mIoU is averaged over the 17 occupied classes, whereas occupancy IoU merges these classes into a single occupied category. Both metrics are computed from the confusion matrix accumulated over all evaluated frames, with zero IoU assigned to classes with zero union. Additional evaluation under the SurroundOcc-nuScenes protocol [44] is provided in supplementary Sec. C.

4.2. Implementation Details

We use frozen DINOv2 [33] and VGGT [42] encoders with metric depth from Metric3D v2 [12]. Their outputs are precomputed, and the appearance neck remains frozen. Images are resized to 504×896 for DINOv2 and padded and resized to 518×518 for VGGT. The model contains 7,200

Method	OccIoU ↑	mIoU ↑	others	barrier	bicycle	bus	car	const. veh.	motorcycle	pedestrian	traffic cone	trailer	truck	drive. surf.	other flat	sidewalk	terrain	manmade	vegetation
MonoScene [4]	29.55	6.06	1.75	7.23	4.26	4.93	9.38	5.67	3.98	3.01	5.90	4.45	7.17	14.91	6.32	7.92	7.43	1.01	7.65
BEVDet [13]	38.23	19.38	4.39	30.31	0.23	32.26	34.47	12.97	10.34	10.36	6.26	8.93	23.65	52.27	24.61	26.06	22.31	15.04	15.10
OccFormer [53]	41.95	21.93	5.94	30.29	12.32	34.40	39.17	14.44	16.45	17.22	9.27	13.90	26.36	50.99	30.96	34.66	22.73	6.76	6.97
BEVStereo [25]	44.33	24.51	5.73	38.41	7.88	38.70	41.20	17.56	17.33	14.69	10.31	16.84	29.62	54.08	28.92	32.68	26.54	18.74	17.49
BEVFormer [26]	50.78	26.88	5.85	37.83	17.87	40.44	42.43	7.36	23.88	21.81	20.98	22.38	30.70	55.35	28.36	36.0	28.06	20.04	17.69
CTF-Occ [40]	51.24	28.53	8.09	39.33	20.56	38.29	42.24	16.93	24.52	**22.72**	**21.05**	22.98	31.11	53.33	33.84	37.98	33.23	20.79	18.0
RenderOcc‡ [34]	48.25	26.11	4.84	31.72	10.72	27.67	26.45	13.87	18.2	17.67	17.84	21.19	23.25	63.2	**36.42**	**46.21**	44.26	19.58	20.72
TPVFormer [14]	50.82	27.83	7.22	38.90	13.67	**40.78**	45.90	17.23	19.99	18.85	14.30	**26.69**	**34.17**	55.65	35.47	37.55	30.70	19.40	16.78
SurroundOcc [44]	53.13	29.21	8.64	40.12	23.36	39.89	45.23	**17.99**	24.91	22.66	18.11	21.64	32.5	57.6	34.10	35.68	32.54	21.27	20.27
GaussianFormer-2† [16]	53.75	29.62	9.19	**40.16**	**23.57**	39.62	45.71	16.97	25.13	20.54	17.83	21.65	32.88	58.27	34.18	40.29	34.77	22.69	20.05
CGOcc (Ours)	**56.32**	**31.10**	**10.53**	37.21	21.80	40.30	**47.82**	17.66	**26.25**	18.82	17.46	22.95	33.90	**63.78**	33.79	43.91	**45.17**	**24.64**	**22.78**

Table 1. **3D semantic occupancy prediction results on the Occ3D-nuScenes validation set.** †GaussianFormer-2 uses the 128-channel configuration (Ch. = 128). ‡RenderOcc denotes the variant trained with both 3D voxel labels and LiDAR-derived 2D semantic and depth labels. Column-wise best results are **bold** and second best results are underlined.

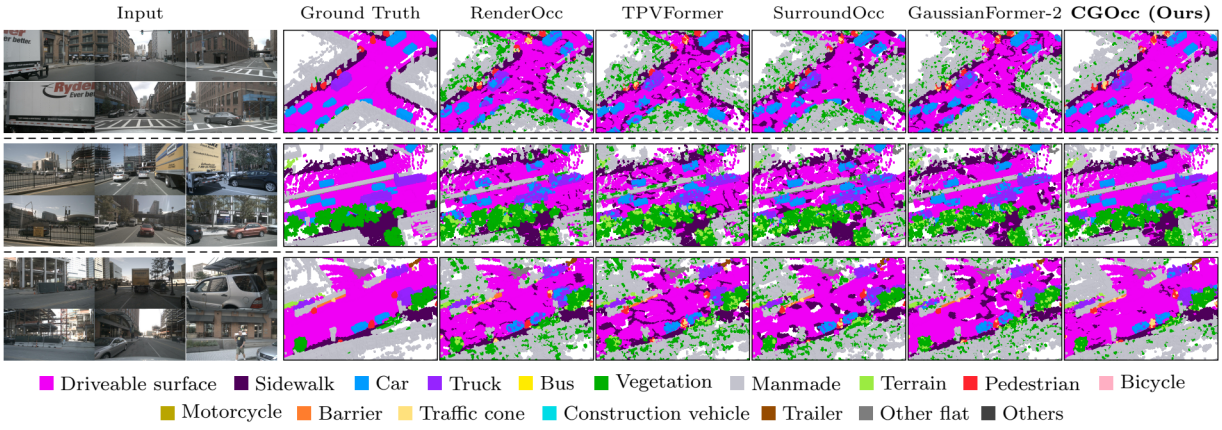


Figure 4. **Qualitative comparison on Occ3D-nuScenes validation set.**

Gaussian candidates and two decoder blocks with 256 hidden dimensions and four attention heads. Query embeddings and the appearance frontend are initialized from the released GaussTR checkpoint [17].

Our main model was trained for 30 epochs on 8 NVIDIA RTX 5090 GPUs with a total batch size of 8. AdamW [29] uses a weight decay of 0.01 and a peak learning rate of 10^{-3}, reduced to 3×10^{-4} for geometry updates and 2×10^{-4} for the dependency encoder and prediction heads. We use a 30-epoch cosine schedule with 1.5 epochs of linear warm-up. Neural blocks use FP16, while geometry, sparse aggregation, and losses remain FP32.

4.3. Main Results

3D semantic occupancy results. Table 1 compares CGOcc with representative vision-based methods on Occ3D-nuScenes. CGOcc achieves 56.32% OccIoU and 31.10% mIoU, the highest scores among the listed methods, exceeding GaussianFormer-2 [16] by 2.57 and 1.48 percentage points, respectively. It also leads in seven of the 17 semantic classes. Figure 4 compares CGOcc with representative baselines, including GaussianFormer-2 and SurroundOcc [44], across three scenes. CGOcc predicts more continuous drivable surfaces with fewer holes and scattered predictions. Vehicle regions have more regular shapes, while roadside vegetation and manmade structures more closely follow the ground-truth spatial layout.

Semantic refinement near boundaries. Figure 5 visualizes local prediction changes from our independently trained diagonal variant CGOcc (Diag), GaussianFormer-2 [16], and SurroundOcc [44] to CGOcc. In the selected crops, CGOcc corrects errors around vehicle contours and occupied/free interfaces, although some regions still exhibit false occupancy and incorrect class assignments. Near semantic interfaces, class-conditional cross-Gaussian statistics may help refine predictions where overlapping prim-

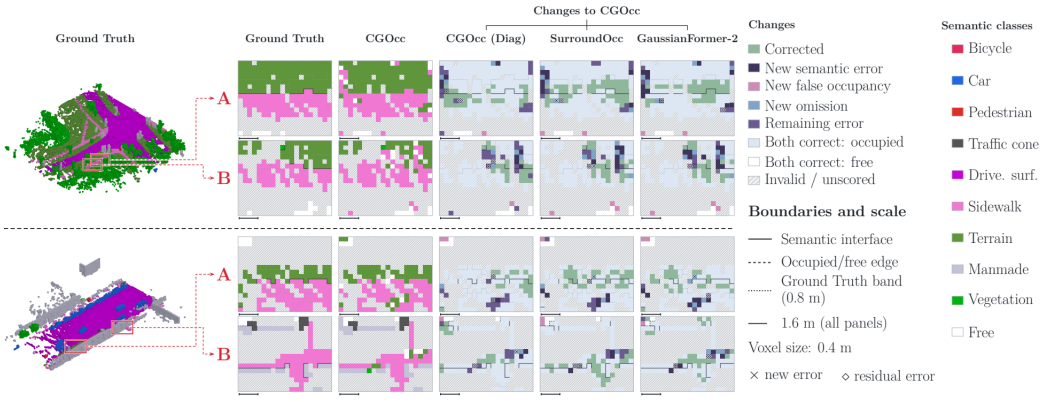


Figure 5. **Visualization of prediction changes near boundaries on Occ3D-nuScenes.** Local maps show changes from each baseline to CGOcc. Corrections occur along semantic interfaces and occupied/free edges, alongside residual and new errors.

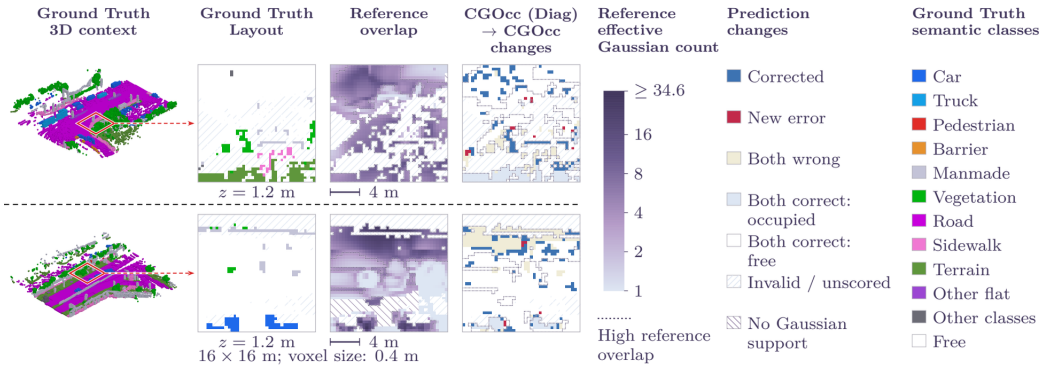


Figure 6. **Visualization of Gaussian overlap and prediction changes from CGOcc (Diag) to CGOcc.** Overlap is measured using a fixed CGOcc (Diag) reference. Blue and red mark corrected predictions and new errors, respectively.

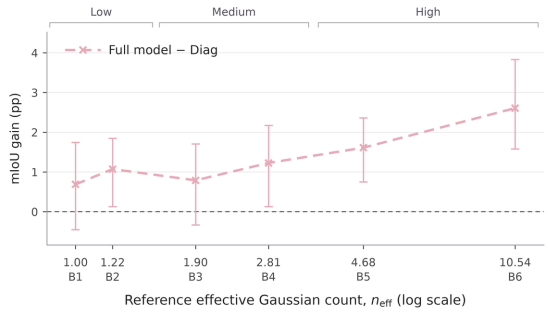


Figure 7. **Performance gains across reference-defined Gaussian overlap levels.** Error bars show 95% confidence intervals from paired scene bootstrap.

Feature sources	DINOv2	VGGT	mIoU (%) ↑	OccIoU (%) ↑
DINOv2 only	✓	×	28.71	54.31
VGGT only	×	✓	30.36	55.22
DINOv2 + VGGT	✓	✓	**31.10**	**56.32**

Table 2. **Ablation of feature sources.**

Initialization depth	mIoU (%) ↑	OccIoU (%) ↑
Uniform z-depth	24.53	51.38
Metric3D v2	**31.10**	**56.32**

Table 3. **Ablation of depth initialization.**

itives provide competing class evidence. Distance-based comparisons beyond these selected crops are provided in the supplementary extended boundary analysis, with boundary definitions and distance binning detailed in supplementary Sec. B.2.

Semantic refinement across Gaussian overlap levels. We compare CGOcc with its diagonal variant across overlap levels defined by the effective Gaussian contribution count of a fixed CGOcc (Diag) reference (supplementary Sec. B.3). Using identical voxel groups for both models, Fig. 7 shows positive mIoU gains over the diagonal variant in every bin. The gains vary across low- and medium-

Decoder variant	Diag	Cov*	S_j
Full (F)	29.42 ± 0.19	31.10 ± 0.20	—
Within-source (W)	29.23 ± 0.20	29.87 ± 0.24	$\mathbf{1.04 \pm 0.23}$
w/o geom. bias (G)	29.11 ± 0.17	30.08 ± 0.24	$\mathbf{0.71 \pm 0.19}$
w/o neighbor msg. (N)	29.25 ± 0.18	29.71 ± 0.22	$\mathbf{1.22 \pm 0.23}$

Table 4. **Decoder–covariance interaction.** $S_j = \Delta_F - \Delta_j$, where $\Delta_j = Y_{j,\mathrm{Cov}} - Y_{j,\mathrm{Diag}}$ and Y denotes mIoU. Larger positive values indicate stronger positive synergy between decoder mechanism j and covariance-aware prediction.* denotes Full model.

overlap bins, then increase at high overlap and peak in the highest bin. This trend suggests that explicitly modeling class-conditional cross-Gaussian interactions can aid semantic refinement when multiple primitives contribute to the same voxel (Sec. 3.3). Figure 6 pairs reference overlap maps with prediction changes from CGOcc (Diag) to CGOcc. In high-overlap regions, CGOcc corrects some errors but leaves others unresolved and introduces errors at voxels correctly classified by the diagonal variant.

4.4. Ablation Study

Ablation of feature sources. We evaluate DINOv2 appearance cues [33] and VGGT multi-view geometric features [42] separately and jointly (Table 2). VGGT alone outperforms DINOv2 alone on both metrics, while their combination performs best. The gain from adding appearance features suggests that they provide semantic evidence complementary to geometric scene structure, supporting their joint aggregation in the multi-source Gaussian interaction module (Sec. 3.2).

Ablation of depth initialization. Replacing depth estimates from Metric3D v2 [12] with uniform z-depth reduces mIoU and OccIoU by 6.57 and 4.94 percentage points, respectively (Table 3). These drops suggest that learned geometry updates do not fully compensate for the loss of scene-dependent depth at initialization.

Decoder–covariance interaction. We examine how decoder mechanisms affect the mIoU advantage of CGOcc over CGOcc (Diag) using eight paired training seeds (Table 4). W restricts observation attention to each backbone; G removes geometry bias from query-attention logits; N removes Gaussian-neighbor messages while retaining the FFN. All inputs and other modules are retained. CGOcc outperforms Diag in every configuration, but each restriction reduces its advantage. These results support positive conditional synergy between decoder mechanisms and covariance-aware readout, with neighbor messages showing the largest mean interaction. By combining observations and local context in Gaussian features, the decoder may help the readout learn useful relationships between primitives.

Readout Variant	mIoU (%) ↑	OccIoU (%) ↑
$\mathbf{P}^{(0)}$-only	29.25	54.56
Diag	29.42	54.78
Class-agnostic	30.28	56.20
Cov (aligned factors)	30.43	56.24
Cov*	**31.10**	**56.32**
Cov (cross off)	30.30	56.13

Table 5. **Ablation of semantic readout.** Variants above the separator are trained independently. Cov (cross off) evaluates the trained Cov checkpoint with only the explicit cross-statistic input set to zero. * denotes Full model.

Ablation of semantic readout. Table 5 compares semantic readout variants for voxel prediction. Diag matches Cov's architecture and parameter count, but its correction uses the diagonal descriptor,

$$\mathbf{r}_v^{\mathrm{Diag}} = H_v(\mathbf{m}_v) - H_v(\mathbf{0}). \tag{11}$$

This descriptor summarizes individual contributions to each class and is insensitive to the relative directions of the contributing factors. Class-agnostic improves on Diag by supplying the normalized form of the cross statistic

$$\mathbf{b}_v = \left(\sum_g w_{vg}\boldsymbol{\ell}_g\right) \odot \left(\sum_g w_{vg}\boldsymbol{\ell}_g\right) - \sum_g w_{vg}^2(\boldsymbol{\ell}_g \odot \boldsymbol{\ell}_g). \tag{12}$$

Here $\odot$ denotes elementwise multiplication. This statistic groups interactions by factor dimension and weights each pair by spatial overlap alone. Cov instead computes a cross statistic for each class c, weighting each pair by $w_{vg}w_{vh}p_{gc}p_{hc}$. Even two strongly overlapping Gaussians therefore contribute little to X_{vc} if either assigns little probability to c.

Cov (aligned factors) retains this class weighting but replaces $\boldsymbol{\ell}_g^\top \boldsymbol{\ell}_h$ with $\|\boldsymbol{\ell}_g\|_2\|\boldsymbol{\ell}_h\|_2$ in X_{vc}. Every pair then makes a nonnegative contribution. In full Cov, pairs with the same class weights and factor norms can make contributions of opposite sign, depending on their relative factor directions. The higher mIoU of Cov suggests that these directions carry information beyond the strength of shared class support.

In Cov (cross off), setting $\boldsymbol{\xi}_v$ to zero the trained Cov checkpoint recovers the learned reference $\mathbf{B}_v$. The resulting is consistent with the correction in Eq. (8), which redistributes probability among occupied classes while preserving their total mass.

5. Conclusion

We present CGOcc for semantic occupancy prediction, combining geometry-conditioned multi-source Gaussian interaction with class-conditional cross-Gaussian readout. The readout refines occupied-class probabilities while preserving the occupied mass and free-space probability of a learned reference. With these designs, CGOcc ahieves state-of-the-art performance on nuScenes while spatial analyses show larger gains at high Gaussian overlap and improved predictions near boundaries. The full semantic readout outperforms its ablated variants, and paired ablations support positive decoder–covariance interactions.

CGOcc still has several limitations, including it relies on calibrated cameras and depth estimates from a pretrained model, and does not explicitly model temporal dependencies. Future work could improve robustness to depth initialization errors and extend cross-Gaussian dependency modeling across frames.

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CGOcc: Learning Cross-Gaussian Dependencies for 3D Semantic Occupancy Prediction

Supplementary Material

A. Additional Methodology Details

A.1. Initialization, Decoder, and Supervision

Fixed image references use bilinearly sampled Metric3D v2 depth for back-projection. Rotations follow the source camera; scales depend on depth, opacity is constant, and read offsets are zero. A learned projection of summed query and slot embeddings initializes queries. Valid initialization requires valid depth and an in-volume center.

Each block samples at the center plus its read offset and at a second location displaced by a $\tanh$-bounded query prediction scaled by λ_{off} times the clipped mean Gaussian scale. Sampling accounts for resizing, padding, and projection validity. Source-level affine projections and pre-normalized residual adapters align features; appearance levels use masked-softmax fusion, whereas VGGT levels remain separate. Camera, source, source-camera-membership, and sampling-point embeddings identify observations. Geometry-bias metadata comprises camera-axis depth, distance, source-camera membership, validity, sampling identity, and learned cues; an encoder and head-specific mapping produce a scaled $\tanh$ bias. Empty observation sets suppress attention but retain the FFN. Only Block 1 uses the local graph: initially valid non-self neighbors send values weighted by attention with learned relative-position bias. Their sum and the unweighted mean of MLP-encoded relative positions normalized by R are jointly projected. All Gaussians may receive messages. Message and FFN residuals share scale η; empty neighborhoods skip both.

Shared heads refine Gaussian states. Center increments accumulate before the bounded update, with $\mathbf{a}_g^{(0)} = \mathbf{0}$:

$$\begin{aligned} \mathbf{a}_g^{(\ell+1)} &= \mathbf{a}_g^{(\ell)} + f_\mu(\mathbf{q}_g^{(\ell+1)}), \\ \boldsymbol{\mu}_g^{(\ell+1)} &= \boldsymbol{\mu}_g^{(0)} + B_g \tanh(\mathbf{a}_g^{(\ell+1)}). \end{aligned} \tag{13}$$

Read offsets, log scales, and opacity logits use bounded increments; rotations use normalized quaternion composition of scaled axis-angle updates. Intermediate labels use floor-based voxel indexing at detached updated centers with valid initialization. Cross-entropy uses in-volume, camera-visible occupied targets, normalized by their class-weight sum; empty sets give zero loss. Terminal auxiliary targets exclude invalid initializations and out-of-volume centers but use no visibility mask; only occupied targets enter semantic loss.

A.2. Dependency Encoding and Readout Heads

Gaussian inputs comprise geometry, opacity, class probabilities, entropy, top-two margin, and valid observations pooled at final centers, with fixed linear projection and normalization. Spatially weighted pooling forms $\mathbf{z}_v$ from encodings, class/factor means and their outer product, mean d_g, semantic variance, $\log(1 + S_v)$, $\log(1 + n_v^{\text{eff}})$, and $\log \mathbf{P}_v^{(0)}$. The reference and correction branches have separate encoders, each using layer normalization and two 64-wide SiLU layers, followed by linear factor and diagonal heads. Occ3D inputs and contexts have 45 and 263 dimensions. Projection in Eq. (5) costs $O(ECr + VCr)$ for E nonzero voxel–Gaussian pairs and V voxels; factor storage is $O(Nr + N)$. The positive-definite kernel permits signed cross terms, which are not determined by diagonal statistics and only couple Gaussians with shared voxel support.

Independent MLPs T_B, T_H use layer normalization and SiLU layers of widths 128 and 64. For $T_v^B(\mathbf{a}, \mathbf{b}) = T_B([\mathbf{z}_v^B; \mathbf{a}; \mathbf{b}])$, the reference adjustments are

$$\begin{aligned} \mathbf{t}_v^B &= T_v^B(\mathbf{m}_v^B, \mathbf{m}_v^B) - T_v^B(\mathbf{0}, \mathbf{0}), \\ \mathbf{s}_v^B &= \beta_s \operatorname{ctr}[\tanh(W_s \mathbf{t}_v^B / \beta_s)], \\ b_v^B &= \beta_o \tanh(\mathbf{w}_o^\top \mathbf{t}_v^B / \beta_o). \end{aligned} \tag{14}$$

These enter Eq. (6). The correction head is

$$H_\psi(\mathbf{z}, \mathbf{m}, \mathbf{t}) = Q W_H T_H([\mathbf{z}; \mathbf{m}; \mathbf{t}]), \tag{15}$$

where Q is the fixed orthonormal Helmert basis for the zero-sum subspace. Both MLPs take 297 inputs on Occ3D; bias-free projections $W_s, \mathbf{w}_o, W_H$ are initialized to zero.

A.3. Readout Controls and Properties

Diag retains both encoders, all factor/diagonal heads, and the full correction head, using Eq. (11). Low-rank self-contributions remain in D_{vc}; L is not zeroed. Its architecture and parameter count match Cov, with all task-specific modules jointly trained. Cross off instead fixes a Cov checkpoint and zeros only $\boldsymbol{\xi}_v$, recovering its own reference $\mathbf{B}_v$.

Class-agnostic normalizes the vector in Eq. (12) using

$$\begin{aligned} D_v^{\text{agn}} &= \sum_g w_{vg}^2 (d_g + \|\boldsymbol{\ell}_g\|_2^2), \\ \eta_{vj} &= \frac{b_{vj}}{D_v^{\text{agn}} + \epsilon_X + |b_{vj}|}. \end{aligned} \tag{16}$$

The full r-vector replaces $\boldsymbol{\xi}_v$ in the shared-head difference; hidden layers are unchanged, while the input dimension changes from $d_z + 2C$ to $d_z + C + r$. Reference, context, and class-conditional diagonal descriptors remain. Independent training uses the same objective, masks, and schedule. Both class weighting and statistic organization change, so this control does not isolate class weighting alone.

Both corrections reuse Eqs. (7) and (8). Zero incremental input cancels the shared-head difference, recovering $\mathbf{B}_v$; $\tanh$ and centering ensure zero-sum corrections bounded by τ. Normalization preserves occupied mass and free probability, but can change occupied/free labels under the $(C+1)$-way argmax. Unsupported, single-contributor, and zero-mass voxels retain the reference; the reference itself retains $\mathbf{P}_v^{(0)}$ without multiple contributors.

A.4. Implementation Settings

Table 6 gives the Occ3D settings shared by Cov and Diag. Learned scalar initializations are marked "init."; other entries are fixed. In meters, valid depth is $(0.1, 51.2]$ and

$$\begin{aligned} s_{g,\tan}^{(0)} &= \mathrm{clip}(0.01 z_g^{(0)}, 0.10, 0.40), \\ B_g &= \mathrm{clip}(0.30 z_g^{(0)}, 0.30, 5.0). \end{aligned} \tag{17}$$

Both tangential axes use this scale; axial scale and opacity start at $0.10\,\mathrm{m}$ and 0.5. Read-offset, log-scale, and opacity-logit increments are bounded by B_g, 1.5, and 2.0; axis-angle updates are scaled by 0.35. Projection scales and sampling mean scales are clipped to $[0.05, 2.5]\,\mathrm{m}$. Cubic splat stencils use three times the largest world-axis standard deviation, rounded up in voxel units, capped at 16 voxels, and clipped to the volume. Heads act only at $S_v > 10^{-12}$ and $n_v^{\mathrm{eff}} > 1 + 10^{-6}$.

Setting	Value
Gaussians / blocks / attention heads	7,200 / 2 / 4
Hidden / attention FFN width	256 / 512
Adapter / geom. bias hidden width	256 / 64
Dropout / sampling points / slots	0 / 2 / 48
Sampling-offset scale λ_{off}	0.5
Graph neighbors K / radius R	16 / 1.6 m
Factor rank r / correction bound τ	8 / 0.5
$d_{\min}$ / ϵ_X / α	0.02 / 0.02 / 10^{-4}
Reference scales β_s / β_o	0.25 / 0.25
Gating γ / geometry bias β (init.)	1.0 / 5.0
Graph η / adapter residual scale (init.)	0.1 / 0.05

Table 6. **Architecture and readout settings on Occ3D-nuScenes.**

B. Boundary and Overlap Evaluation Protocols

B.1. Evaluation Setup and Metrics

The regional analyses use the validation set from Occ3D-nuScenes dataset [40]. Let $y(v)$ denote the label of voxel v, $\mathcal{K} = \{0, \ldots, 16\}$ the occupied classes, and $f = 17$ free space. Write $\mathcal{Y} = \mathcal{K} \cup \{f\}$. With camera-visibility mask M, the scoring domain is $\mathcal{V} = \{v : M(v) = 1,\ y(v) \in \mathcal{Y}\}$. All methods in a regional comparison share the same target frames, labels, scoring mask, and region assignments. Diag and Cov are separately trained models. Cross off instead uses the same Cov checkpoint and upstream computation, setting only the explicit cross-statistic input to zero while retaining geometry, context, and diagonal statistics.

For model m and region $\mathcal{R}_t \subseteq \mathcal{V}_t$ in frame t, we first accumulate an 18×18 confusion matrix:

$$H_{ij}^{(m,\mathcal{R})} = \sum_t \sum_{v \in \mathcal{R}_t} a_{t,ij}^{(m)}(v), \tag{18}$$

where $a_{t,ij}^{(m)}(v) = 1$ when $y_t(v) = i$ and $\hat{y}_{m,t}(v) = j$, and zero otherwise. Rows denote ground truth and columns predictions. Omitting the superscripts, the class union U_k and semantic mIoU are computed as

$$\begin{aligned} U_k &= \sum_j H_{kj} + \sum_i H_{ik} - H_{kk}, \\ \mathrm{mIoU}(H) &= \frac{100}{17} \sum_{k=0}^{16} \frac{H_{kk}}{U_k}, \end{aligned} \tag{19}$$

where a zero-union class contributes zero. Free space is excluded from the class average but retained in the confusion matrix, so false occupied predictions on free voxels contribute to semantic false positives. OccIoU merges all occupied classes:

$$\mathrm{OccIoU}(H) = \frac{100\,TP}{TP + FP + FN}. \tag{20}$$

A zero occupancy union is undefined in the boundary-distance evaluation. Metrics are computed after pooling counts, rather than averaging frame-level or regional IoUs.

B.2. Boundary Definition and Distance Binning

Let L be the LiDAR-validity mask and $\mathcal{N}_6(v)$ the six face-sharing neighbors of v. The seed set $\mathcal{S}$ contains both endpoints of every pair with $u \in \mathcal{N}_6(v)$, $L(v) = L(u) = 1$, and $y(v) \neq y(u)$. Define Boundary as $\mathcal{B} = \mathcal{V} \cap \mathcal{S}$ and Interior as $\mathcal{I} = \mathcal{V} \setminus \mathcal{S}$. The seed set $\mathcal{S}$ includes interfaces between different occupied classes and between occupied and free voxels, without additional dilation. LiDAR validity determines whether a label transition is confirmed; camera visibility determines scoring. Boundary $\mathcal{B}$ and Interior

$\mathcal{I}$ partition $\mathcal{V}$. Interior includes occupied and free voxels outside the confirmed boundary, including voxels adjacent to unknown regions. These regions depend on ground truth and masks, not model outputs.

For voxel size $h = 0.4\,\mathrm{m}$, the unsigned 3D distance is

$$d(v) = h \min_{u \in \mathcal{S}} \|v - u\|_2, \tag{21}$$

with display neighborhood $\mathcal{N}_{0.8} = \{v \in \mathcal{V} : d(v) \leq 0.8\,\mathrm{m}\}$. If $\mathcal{S}$ is empty, $d(v) = +\infty$. Distances are computed on the full volume before slicing or cropping. The $0.8\,\mathrm{m}$ neighborhood provides display context and does not replace $\mathcal{B}$; it contains some Interior voxels. Distance curves use the bin index

$$\nu(v) = \lfloor d(v)/h + 1/2 \rfloor, \tag{22}$$

with plotted centers $0, 0.4, \ldots, 4.0\,\mathrm{m}$. Distances at least $4.2\,\mathrm{m}$ are retained in a separate tail bin. The zero-distance bin contains precisely the scored boundary seeds.

B.3. Gaussian Overlap Measurement and Binning

Overlap is measured using a fixed Diag reference. For its spatial contributions ω_{vg}, define

$$\begin{aligned} S_v &= \sum_g \omega_{vg}, \qquad w_{vg} = \omega_{vg}/S_v, \\ n_v^{\mathrm{eff}} &= \left(\sum_g w_{vg}^2 \right)^{-1}. \end{aligned} \tag{23}$$

We form overlap groups on $\mathcal{V}^+ = \{v \in \mathcal{V} : S_v > 10^{-12}\}$, including supported free voxels; unsupported voxels are recorded separately. The effective count measures the spread of Gaussian contributions, not Gaussian-center density or total weight.

We pool n_v^{eff} over all supported valid voxels in the entire training set and estimate thresholds using linearly interpolated empirical quantiles:

$$q_j = Q_{j/6}(n^{\mathrm{eff}}), \qquad j = 1, \ldots, 5. \tag{24}$$

With $q_0 = -\infty$ and $q_6 = +\infty$, group $\mathcal{O}_j$ contains voxels $v \in \mathcal{V}^+$ satisfying $q_{j-1} < n_v^{\mathrm{eff}} \leq q_j$. Threshold ties enter the lower group. Low, Medium, and High combine group pairs $(1, 2)$, $(3, 4)$, and $(5, 6)$, respectively. All models use these fixed assignments. Curve positions show each group's pooled median n_v^{eff} on the evaluation set.

B.4. Confidence Intervals via Paired Scene Bootstrap

For overlap group j, let $H_{s,j}^{(m)}$ contain all evaluated frames from scene s. The same scene multiplicities $W_s^{(r)}$ are shared by every model and group:

$$\begin{aligned} H_j^{*(m,r)} &= \sum_{s=1}^{150} W_s^{(r)} H_{s,j}^{(m)}, \\ \Delta_j^{*(r)} &= T(H_j^{*(\mathrm{Cov},r)}) - T(H_j^{*(Diag,r)}), \end{aligned} \tag{25}$$

where T is mIoU in percent. We recompute mIoU from each resampled confusion matrix. Writing $\mathcal{D}_j = \{\Delta_j^{*(r)}\}_{r=1}^{1000}$, the interval is

$$\mathrm{CI}_{95\%,j} = [Q_{0.025}(\mathcal{D}_j), Q_{0.975}(\mathcal{D}_j)]. \tag{26}$$

Gains and intervals are in percentage points. These pointwise percentile intervals describe scene-sampling uncertainty for fixed models and thresholds.

B.5. Visualization Settings and Error Definitions

Corresponding panels share slice coordinates, crop windows, voxel size, and visibility masks. Each local cell represents one voxel; invalid locations are masked rather than treated as free space. Boundary interfaces and distance fields are computed before cropping, so crop edges do not create artificial boundaries. Solid and dashed lines mark occupied-class and occupied/free interfaces, respectively; dotted lines mark the $0.8\,\mathrm{m}$ neighborhood. Local change maps cover all scored voxels in the crop, including its non-boundary portion.

For comparison labels $b(v)$ and Cov labels $c(v)$, Corrected means $b(v) \neq y(v)$ and $c(v) = y(v)$. The two-color overlap maps mark New error only when $b(v) = y(v)$ and $c(v) \neq y(v)$. The detailed boundary maps distinguish new semantic errors, which change a correct occupied class to another occupied class, and new false occupancy, which changes correct free space to occupied. A new omission requires $y(v) \in \mathcal{K}$, $b(v) \in \mathcal{K}$, and $c(v) = f$; the earlier occupied label need not be semantically correct. Remaining error contains Cov errors outside these three categories; both-correct voxels are separated into occupied and free.

Overlap heatmaps use the common logarithmic mapping

$$t(v) = \mathrm{clip}_{[0,1]} \left(\frac{\log n_v^{\mathrm{eff}}}{\log U} \right), \tag{27}$$

with $U \approx 34.58565$, the fixed training-calibration 99th percentile. Darker purple denotes greater overlap; dotted purple contours enclose the High group. Quantitative gain curves connect measured group values by straight segments

without smoothing. Spatial heatmaps and contours carry no confidence intervals. Correction counts describe changes in label accuracy and do not, by themselves, establish an mIoU gain.

C. Extended Experimental Results

We report quantitative and qualitative results under the SurroundOcc-nuScenes evaluation protocol (Table 7 and Fig. 8), followed by extended boundary and Gaussian-overlap analyses under the Occ3D-nuScenes evaluation protocol (Figs. 10 and 11).

SurroundOcc evaluation protocol. For Table 7 and Fig. 8, we use the dense semantic occupancy annotations released by SurroundOcc [44] for nuScenes. Each input comprises RGB images from six surround-view cameras. The prediction volume spans $[-50, 50]$ m along both horizontal axes and $[-5, 3]$ m vertically, discretized into a $200 \times 200 \times 16$ grid with a voxel size of 0.5 m. The label space contains 16 occupied semantic classes and one free-space class. Following the official evaluation protocol, voxels assigned the ignore label are excluded, and no additional camera-visibility mask is applied. We report mIoU averaged over the 16 occupied classes and OccIoU obtained by merging these classes into a single occupied category. Both metrics are computed from intersection and union counts accumulated over all evaluated frames.

Quantitative and qualitative results on SurroundOcc-nuScenes. Table 7 summarizes the quantitative results. CGOcc achieves 21.25% mIoU and 33.65% OccIoU. It obtains the highest aggregate scores among the listed methods, exceeding SurroundOcc [44] by 0.95 percentage points in mIoU and 2.16 percentage points in OccIoU. Figure 8 complements these quantitative results by comparing CGOcc with TPVFormer [14], OccFormer [53], GaussianFormer-2 [16], and SurroundOcc across five validation scenes. In the first two rows, CGOcc predicts more continuous drivable surfaces, while several baselines leave larger gaps across the road. The fourth row shows the elongated bus region and its surrounding terrain and vegetation. In the final row, CGOcc leaves fewer interruptions in the main drivable strip than TPVFormer and GaussianFormer-2. Local gaps and incorrect class assignments remain visible around vehicles and roadside structures. These examples complement the aggregate and per-class results in Table 7.

Extended boundary analysis. We extend the boundary analysis in Sec. 4.3 with distance-based evaluation and additional qualitative comparisons under the Occ3D-nuScenes protocol [40]. Figure 9 compares all evaluated methods using unsigned 3D Euclidean distances to ground-truth boundaries, including both occupied-class and occupied/free interfaces. Among the compared methods, CGOcc achieves the highest mIoU at 0.4 m and the highest OccIoU from 0.4 to 4.0 m. CGOcc outperforms CGOcc (Diag) in both metrics throughout the plotted range; the mIoU margin peaks at 0.4 m and generally narrows with distance. At zero distance, TPVFormer and RenderOcc retain higher mIoU and OccIoU, respectively. GaussianFormer-2 also retains higher mIoU than CGOcc in several bins beyond 0.4 m.

Figure 10 complements these quantitative results with two local regions from each of five scenes, comparing CGOcc with CGOcc (Diag) and four external baselines. Corrected voxels appear along both semantic interfaces and occupied/free edges. Relative to SurroundOcc and GaussianFormer-2, CGOcc corrects some errors at transitions between road, sidewalk, and terrain in the first scene. In the final scene, both comparisons show corrections around vehicle outlines and along the narrow road regions. Residual class errors and new omissions remain alongside these corrections, showing that the local improvements are incomplete.

Extended Gaussian-overlap analysis. We extend the quantitative analysis of semantic refinement across Gaussian overlap levels in Sec. 4.3 with five additional qualitative examples under the Occ3D-nuScenes protocol. Figure 11 pairs ground-truth layouts with reference overlap maps and Diag-to-Cov prediction changes for these scenes. The overlap maps use the effective Gaussian contribution count from the fixed Diag reference (Sec. B.3). Corrections occur within and around the high-overlap contours, including regions near vehicles and manmade structures. These regions also contain new errors and voxels misclassified by both variants, so high reference overlap does not imply that a voxel is corrected. The corresponding evaluation over fixed voxel groups in Fig. 7 shows larger mIoU gains in the high-overlap bins.

Method	OccIoU ↑	mIoU ↑	barrier	bicycle	bus	car	const. veh.	motorcycle	pedestrian	traffic cone	trailer	truck	drive. surf.	other flat	sidewalk	terrain	manmade	vegetation
MonoScene [4]	23.96	7.31	4.03	0.35	8.00	8.04	2.90	0.28	1.16	0.67	4.01	4.35	27.72	5.20	15.13	11.29	9.03	14.86
Atlas [31]	28.66	15.00	10.64	5.68	19.66	24.94	8.90	8.84	6.47	3.28	10.42	16.21	34.86	15.46	21.89	20.95	11.21	20.54
BEVFormer [26]	30.50	16.75	14.22	6.58	23.46	28.28	8.66	10.77	6.64	4.05	11.20	17.78	37.28	18.00	22.88	22.17	13.80	22.21
TPVFormer-lidar [14]	11.51	11.66	16.14	7.17	22.63	17.13	8.83	11.39	10.46	8.23	9.43	17.02	8.07	13.64	13.85	10.34	4.90	7.37
TPVFormer [14]	30.86	17.10	15.96	5.31	23.86	27.32	9.79	8.74	7.09	5.20	10.97	19.22	38.87	21.25	24.26	23.15	11.73	20.81
OccFormer [53]	31.39	19.03	18.65	10.41	23.92	30.29	10.31	14.19	13.59	10.13	12.49	20.77	38.78	19.79	24.19	22.21	13.48	21.35
GaussianFormer [15]	29.83	19.10	19.52	11.26	26.11	29.78	10.47	13.83	12.58	8.67	12.74	21.57	39.63	23.28	24.46	22.99	9.59	19.12
GaussianFormer-2† [16]	30.56	20.02	20.15	**12.99**	27.61	30.23	11.19	**15.31**	12.64	9.63	13.31	22.26	39.68	23.47	**25.62**	23.20	12.25	20.73
SurroundOcc [44]	31.49	20.30	**20.59**	11.68	28.06	30.86	10.70	15.14	**14.09**	**12.06**	14.38	22.26	37.29	**23.70**	24.49	22.77	**14.89**	21.86
CGOcc (Ours)	**33.65**	**21.25**	20.24	12.76	**30.96**	**33.80**	**11.45**	15.26	12.23	10.03	**15.69**	**24.79**	**40.30**	23.57	25.28	**25.42**	14.73	**23.46**

Table 7. **3D semantic occupancy prediction results on the SurroundOcc-nuScenes validation set.** †GaussianFormer-2 uses the 128-channel configuration (Ch. = 128). Column-wise best results are **bold** and second best results are underlined.

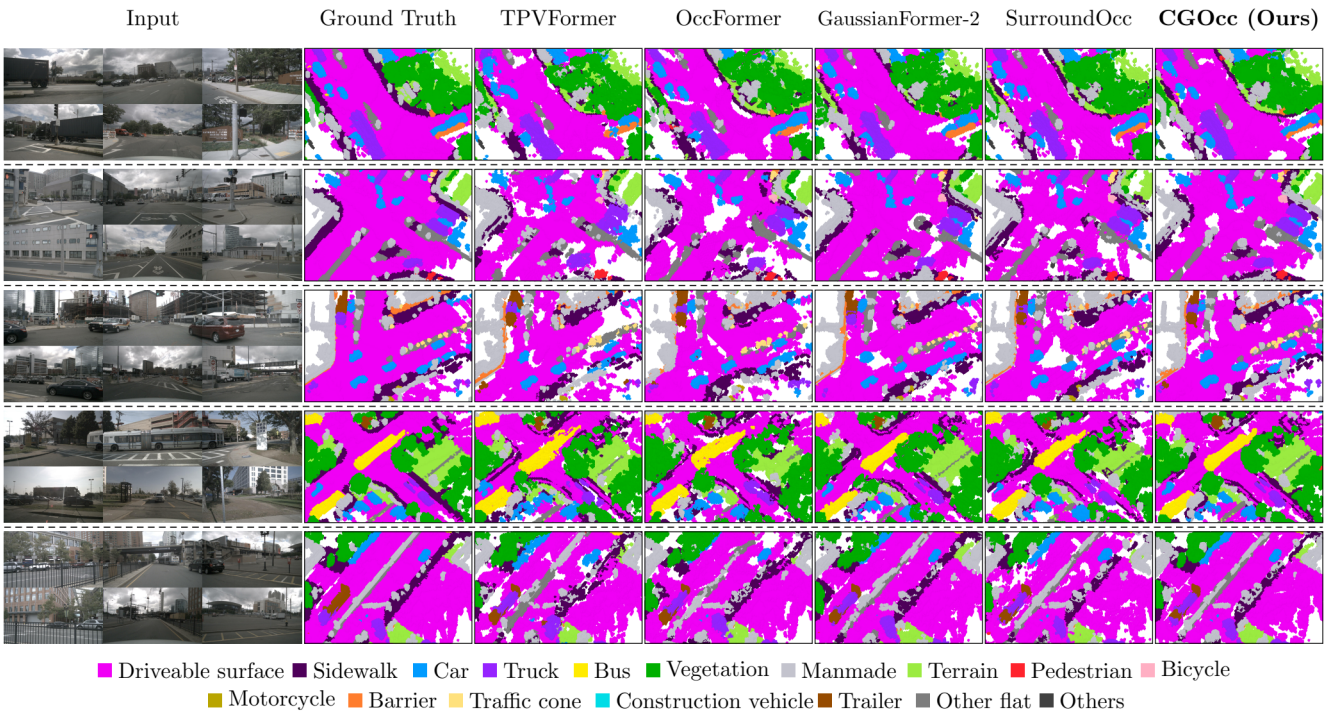
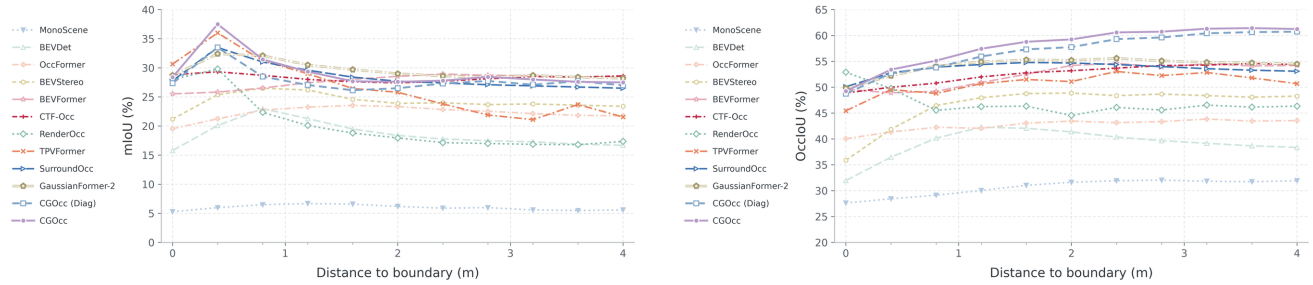


Figure 8. **Qualitative comparison on the SurroundOcc-nuScenes validation set.** Each row shows one of five validation scenes, with surround-view inputs, ground truth, and predictions from representative methods. The examples show differences in drivable-surface continuity and the predicted shapes and semantics of vehicles and roadside structures.



(a) mIoU (b) OccIoU

Figure 9. **Quantitative comparison across distances to ground-truth boundaries.** Panels (a) and (b) report mIoU and OccIoU over common distance bins. Distances are unsigned 3D Euclidean distances to occupied-class and occupied/free interfaces.

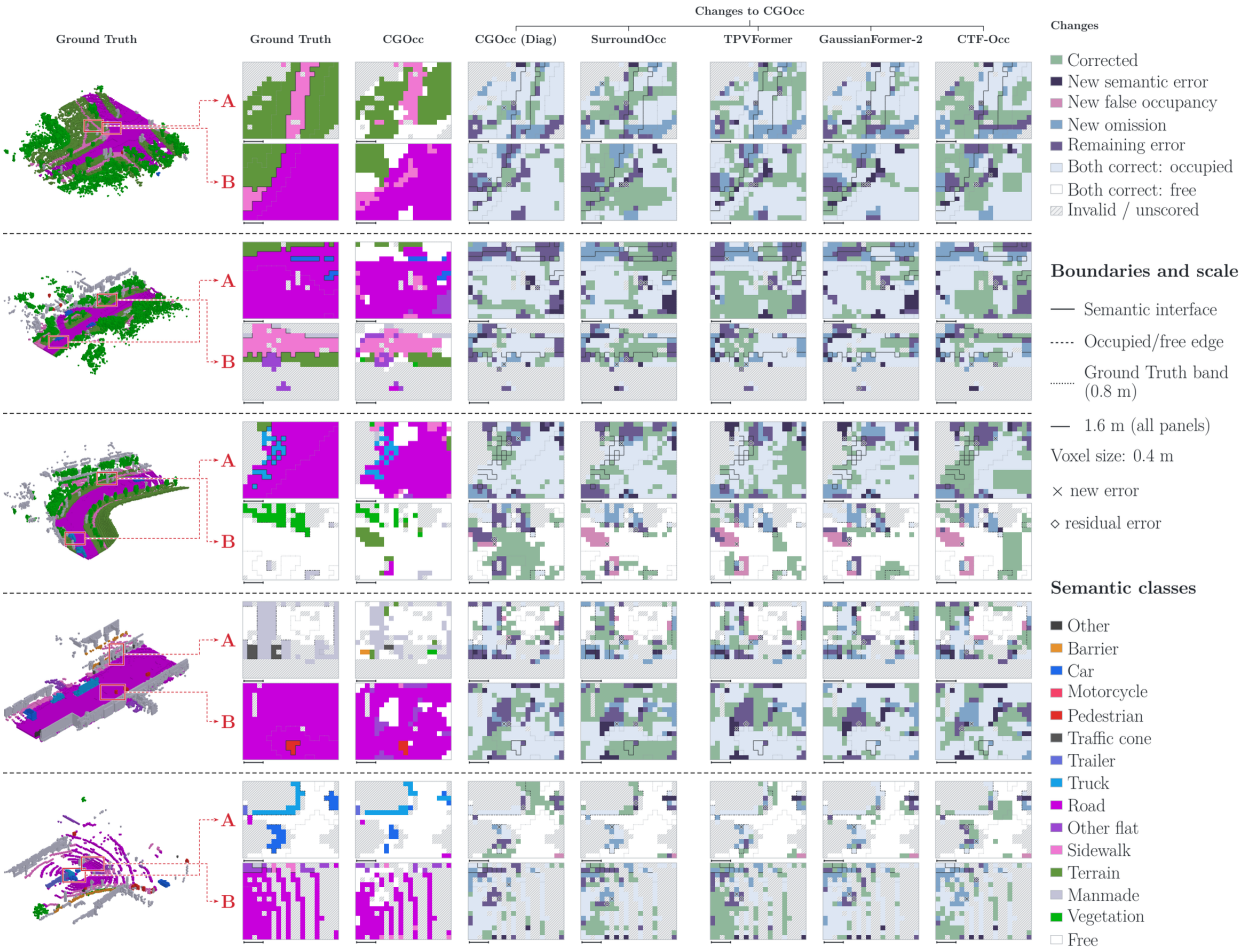


Figure 10. **Visualization of prediction changes near boundaries on Occ3D-nuScenes.** Each scene shows two local regions, with change maps comparing CGOcc against CGOcc (Diag) and external baselines. Solid and dashed lines mark occupied-class and occupied/free interfaces, respectively; dotted lines indicate the 0.8 m boundary neighborhood. Colors distinguish corrected voxels and different error types.

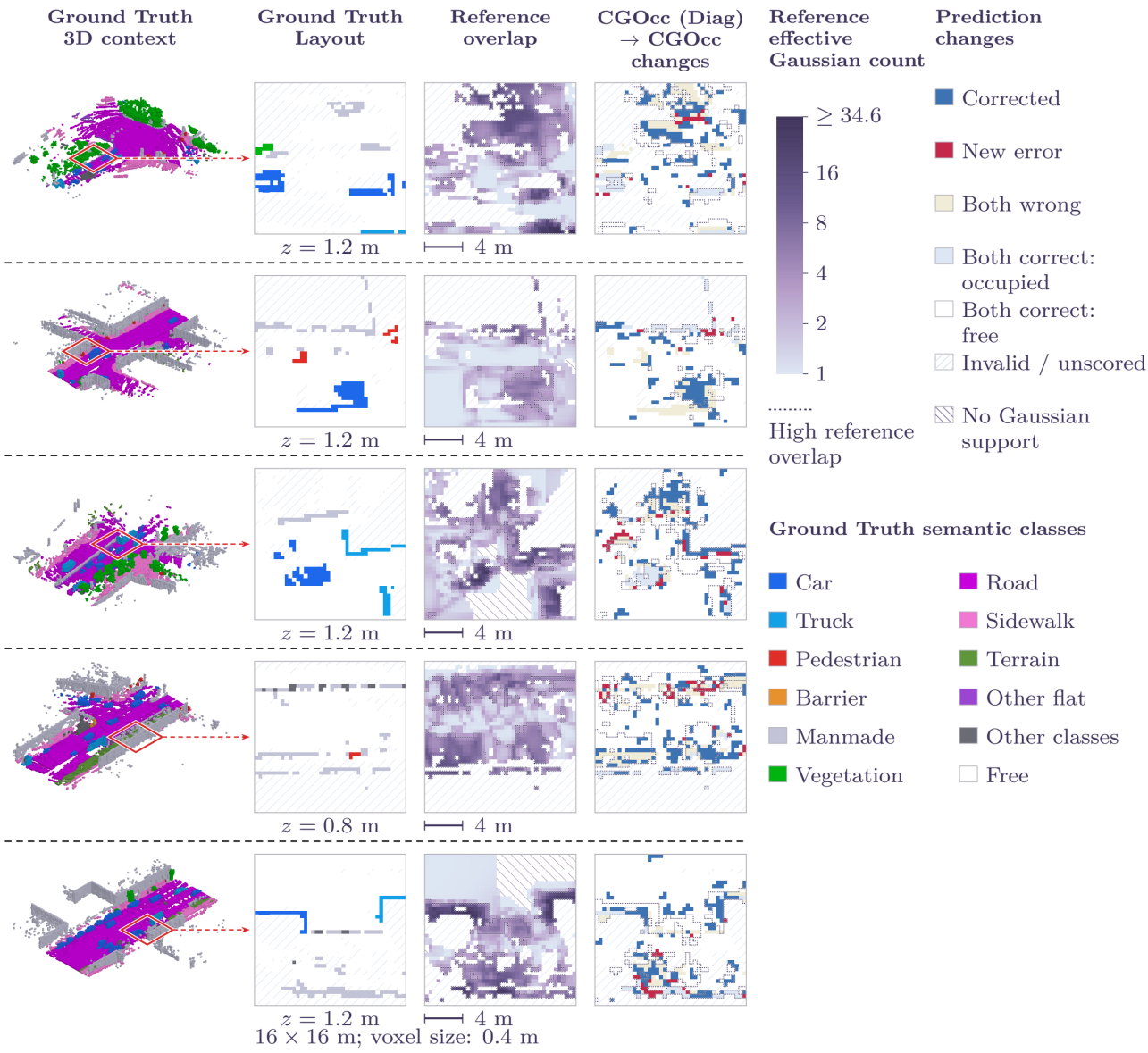


Figure 11. **Visualization of Gaussian overlap and prediction changes from CGOcc (Diag) to CGOcc on Occ3D-nuScenes.** Overlap maps use the effective Gaussian contribution count from a fixed CGOcc (Diag) reference. Darker purple indicates greater overlap, and dotted contours mark high-overlap regions. In the change maps, blue and red denote corrected predictions and new errors, respectively.